Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 06



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Binary Search Tree
- Topic — Construction of BST
- Topic — Search
- Topic — Min
- Topic — Max

Slide

Topics to be Covered



- Topic — BST Deletion
- Topic — Counting
- Topic — AVL tree (height balanced BST)
- Topic
- Topic

Slide



Topic : Tree

BST Counting
AVL tree insertion & rotation

Slide

Question

A binary search tree contains the value 1, 2, 3, 4, 5, 6, 7, 8. The tree is traversed in pre-order and the value are printed out. Which of the following sequence is a valid output?

(a) 5 3 1 2 4 7 8 6 ✗

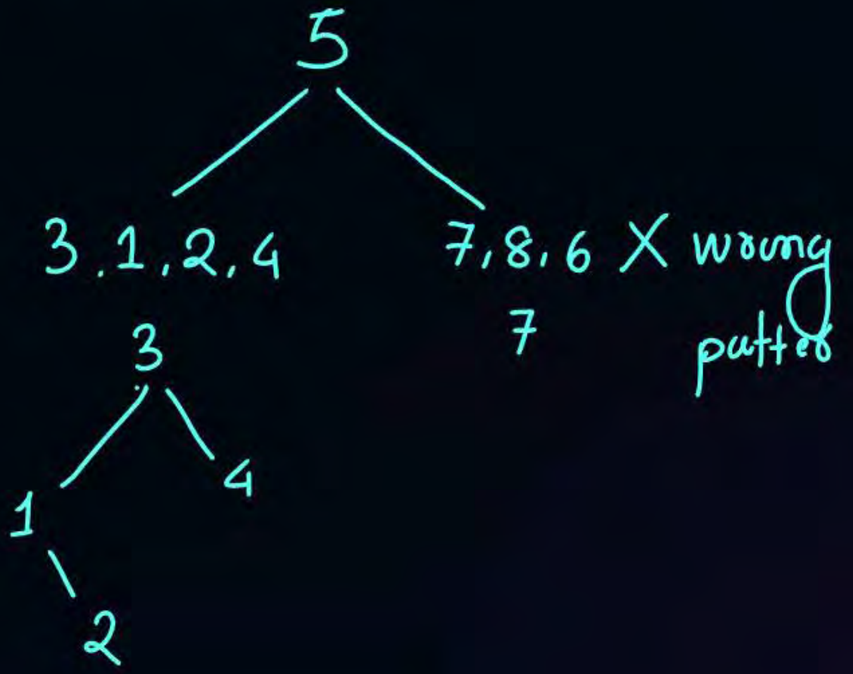
(b) 5 3 1 2 6 4 8 7

(c) 5 3 2 4 1 6 7 8

(d) 5 3 1 2 4 7 6 8

Root Left Right

5 [ ] [ ]





Question

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(a) 5 3 1 2 4 7 8 6 ✗

(b) 5 3 1 2 6 4 8 7 ✗

(c) 5 3 2 4 1 6 7 8

(d) 5 3 1 2 4 7 6 8

Root Left Right

5 [ ] [ ]

Root < 5 > 5

cant divide in the given pattern



Question

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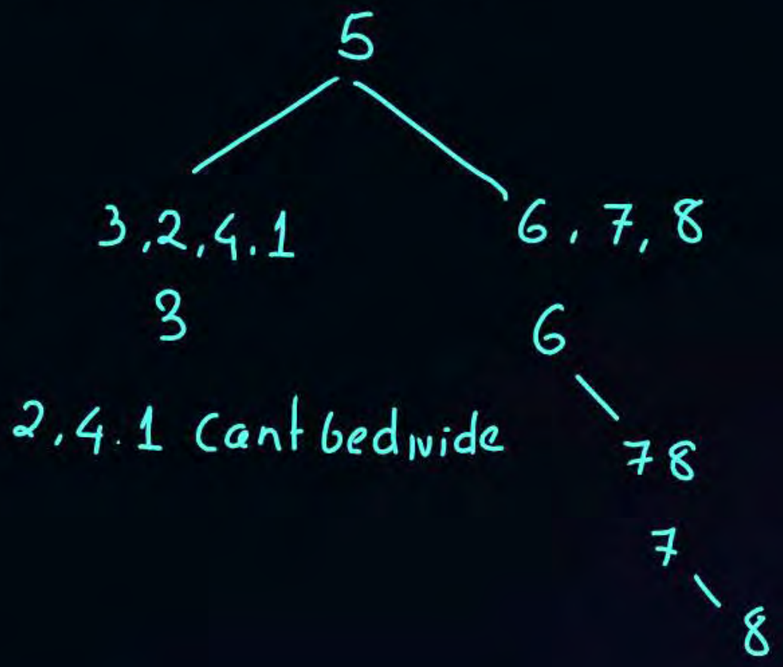
(b) 5 3 1 2 6 4 8 7 ✗

(c) 5 3 2 4 1 6 7 8

(d) 5 3 1 2 4 7 6 8

Root Left Right

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Question

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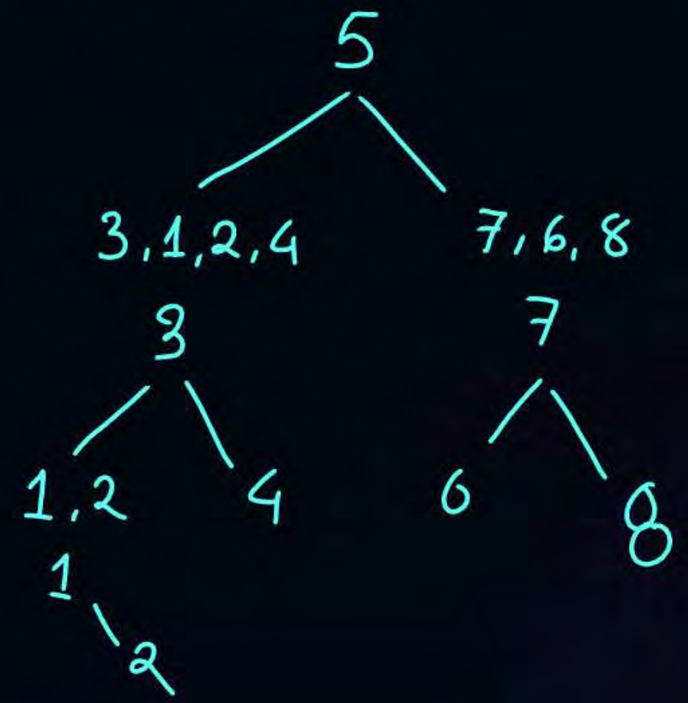
(b) 5 3 1 2 6 4 8 7 ✗

(c) 5 3 2 4 1 6 7 8

(d) 5 3 1 2 4 7 6 8

Root Left Right

5 [ ] [ ]





Question

#Q The numbers 1, 2, ...n are inserted in a binary search tree in some order. In the resulting tree, the right subtree of the root contains p nodes. The first number to be inserted in the tree must be.

(A) p ✗

(B) p + 1 ✗

(C) n - p ✓

(D) n - p + 1 ✗

n-p

1, 2, 3, 4, 5, 6, 7

8, 9, 10



n-p-1 — p



Topic : Binary Search Tree

Inorder predecessor

Inorder : 3, 6, 7, 8, 11, 17, 26, 28

predecessor of an element x is

an element that occurs before

x in inorder Sequence

Minimum element predecessor does not exits



Slide



Topic : Binary Search Tree

Inorder Successor

11 Successor is 17
26 Successor is 28

Inorder : 3, 6, 7, 8, 11, 17, 26, 28

Successor of an element that occurs after an element in inorder Sequence

No Successor of Max element



Slide



Topic : Binary Deletion

Delete node *x* from a tree *T*

We can distinguish three cases

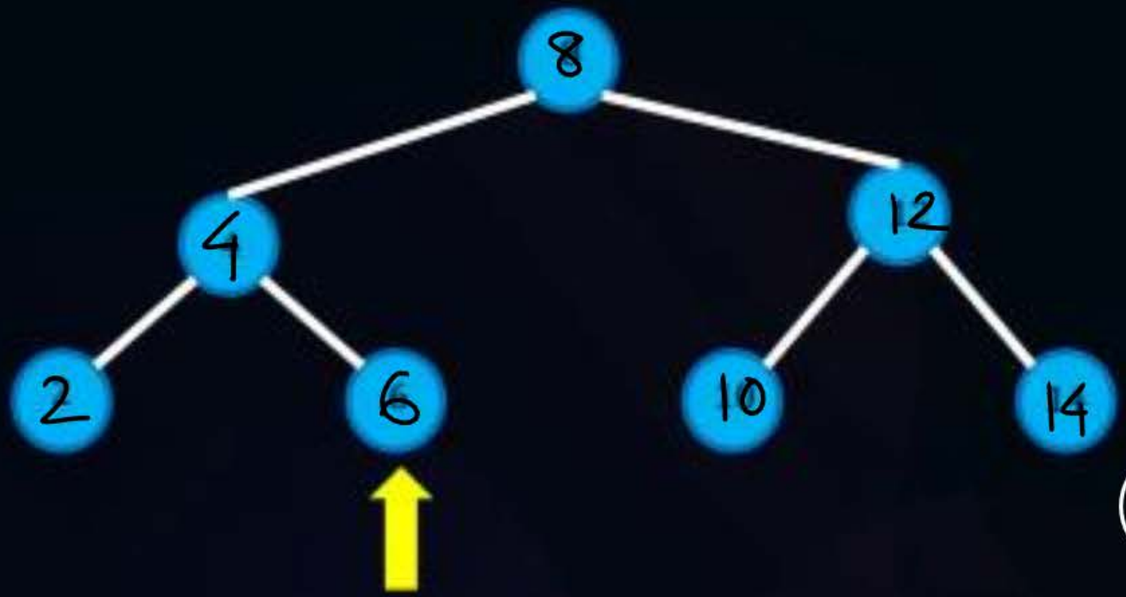
x has *no* children

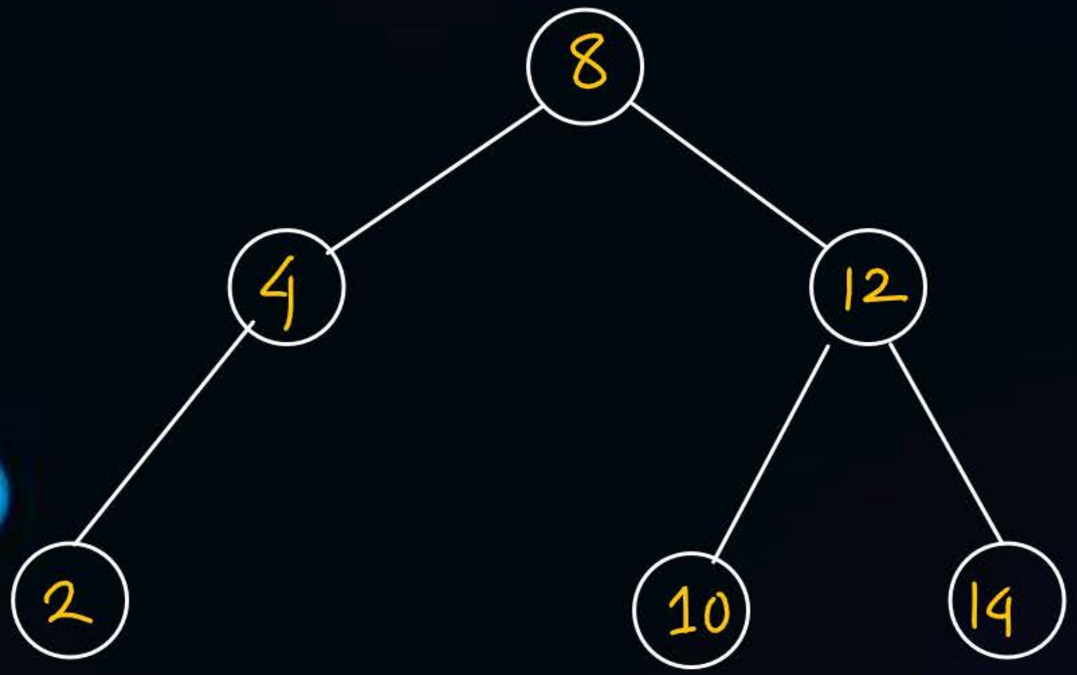
x has *one* child

x has *two* children ⟶ element x replaced by inorder predecessor or inorder Successor

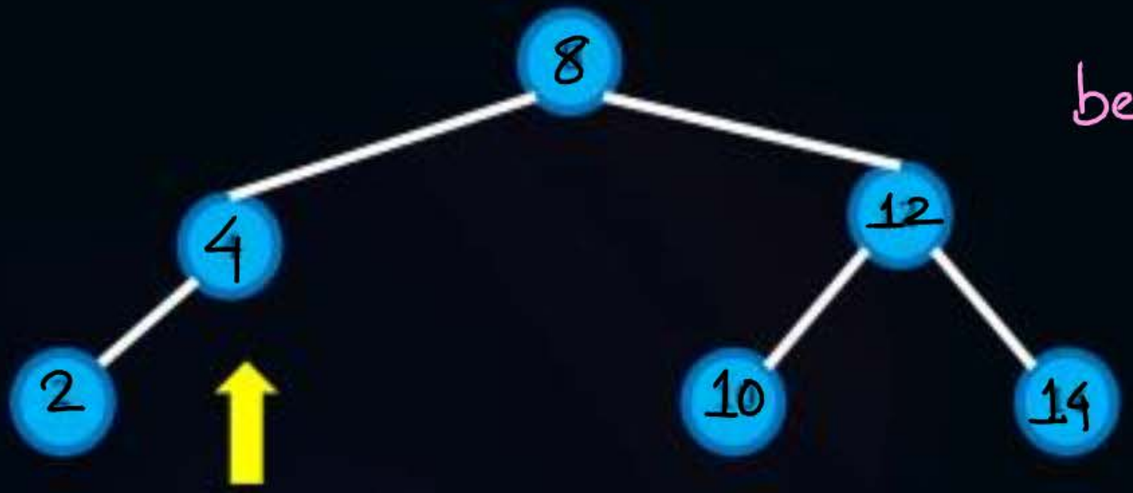
Deletion Case 1

If *x* has no children – just remove *x*





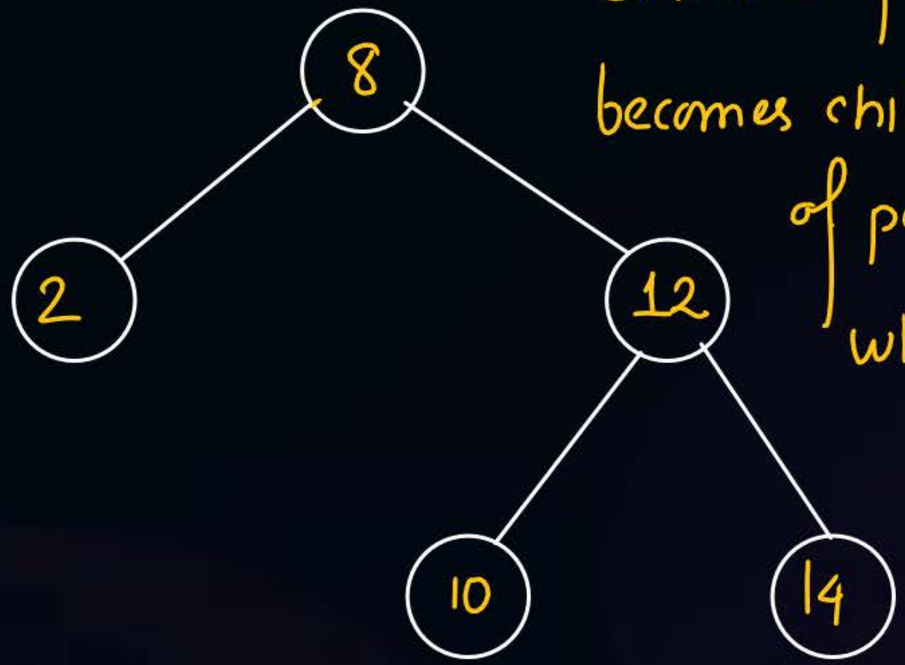
Deletion Case 2



if a Node is having Single child

and we delete then children of x

becomes children of parent of x



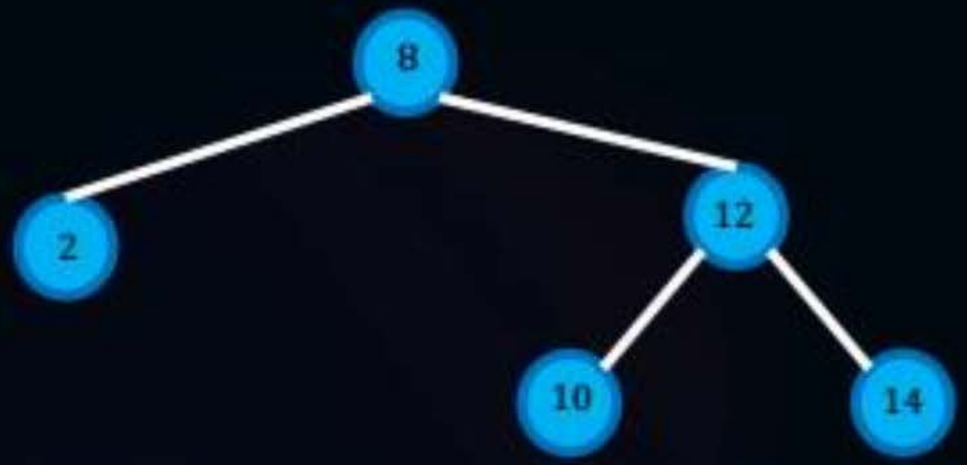
children of 4

becomes children

of parent of 4

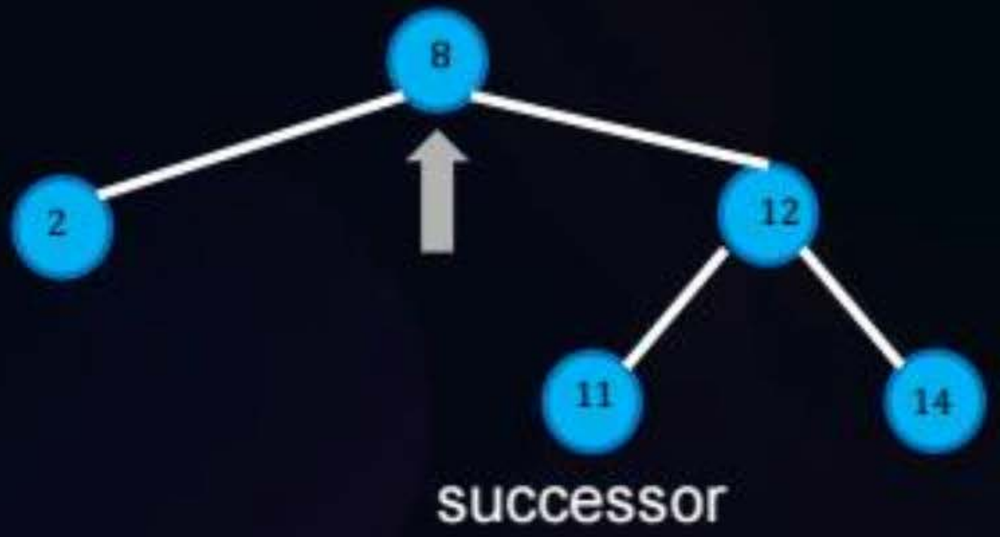
which is 8

Deletion Case 2

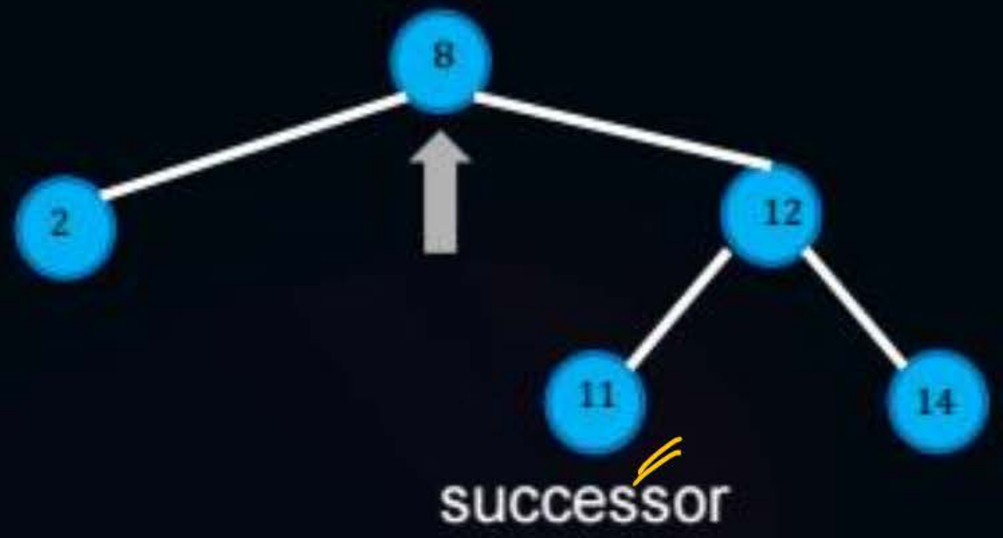




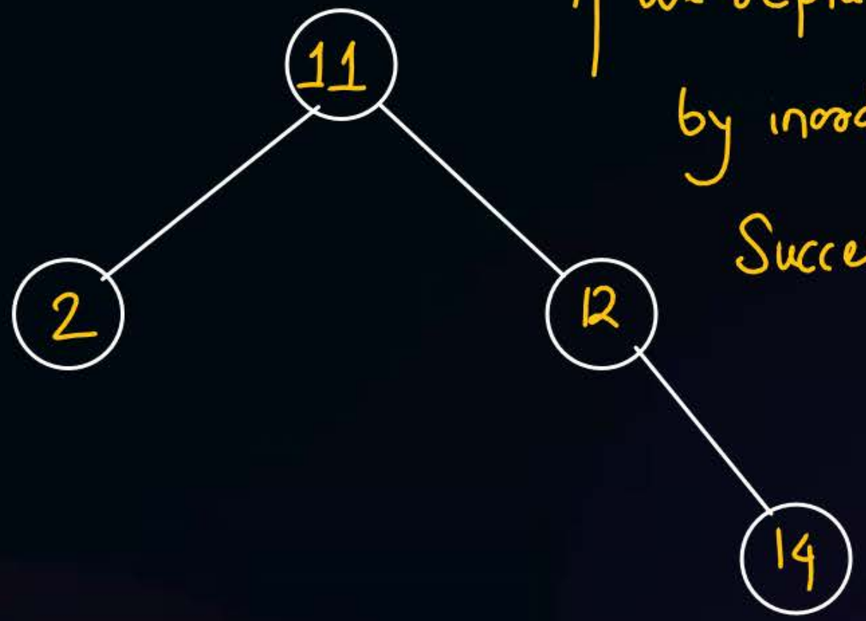
Topic : Deletion Case 3



Deletion Case 3



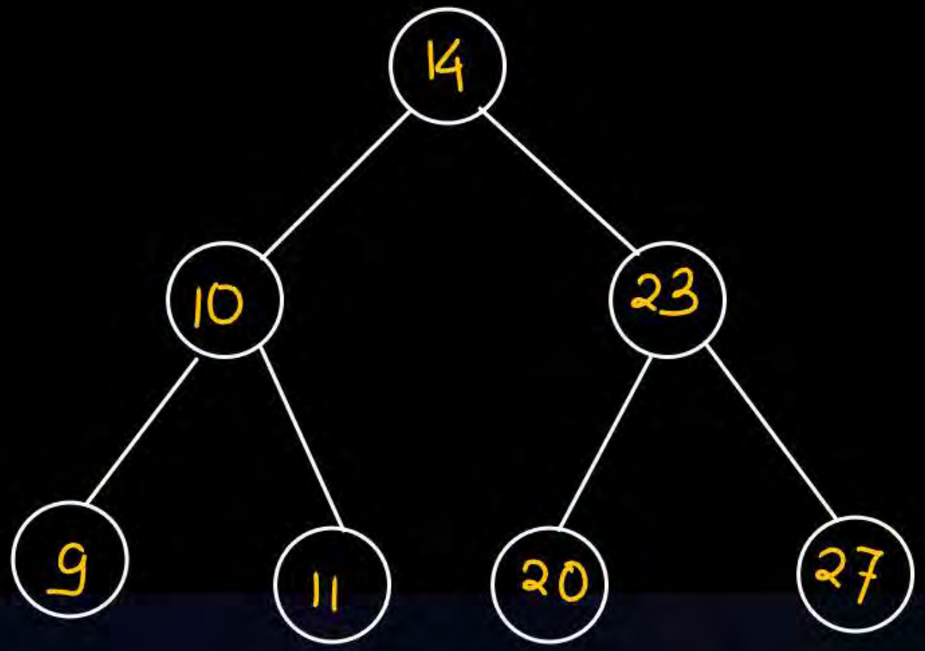
if we delete a Node having 2 children
then we replace x by processor of x
or Successor of x.



if we replace by inorder Successor

Deletion Case 3

How many children a Successor/predecessor can have



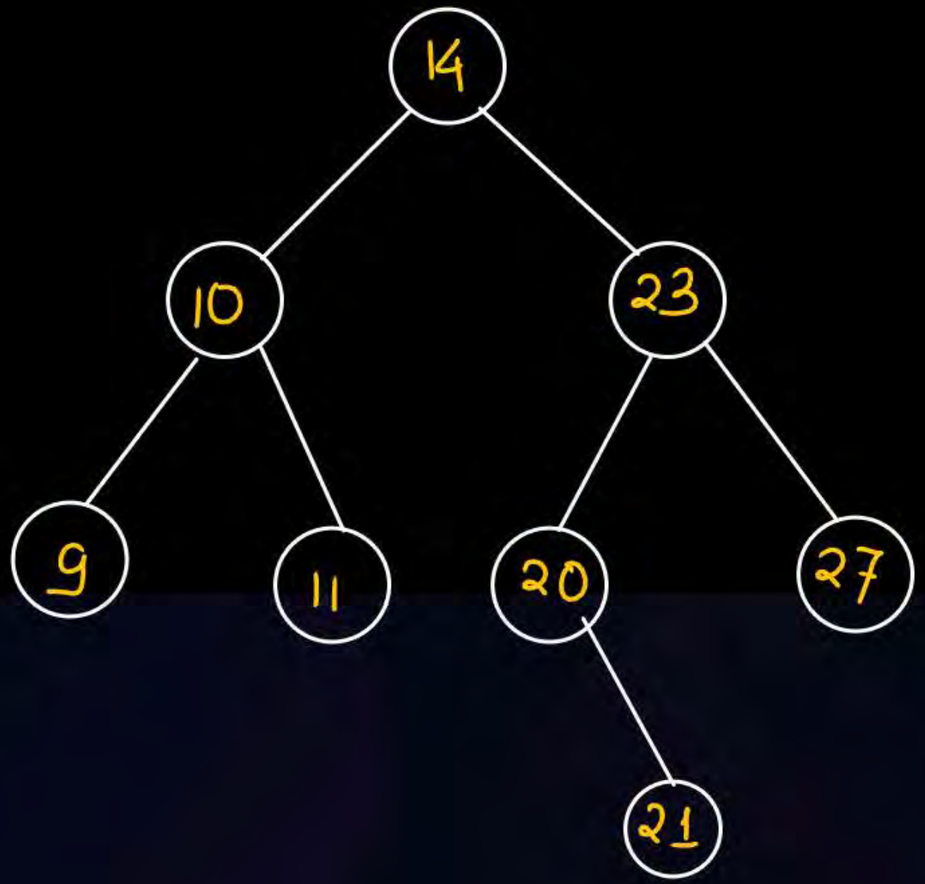
14 predecessor 11

14 Successor 20

0 - children

Deletion Case 3

How many children a successor/predecessor can have



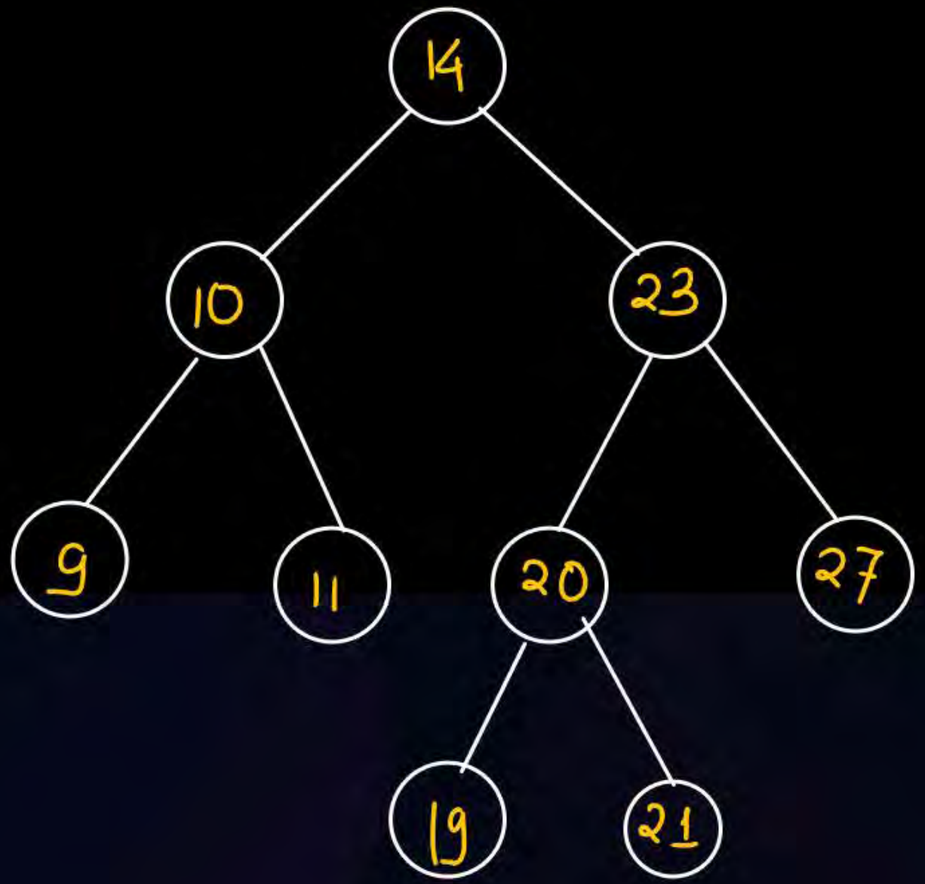
14 predecessor - 11

14 Successor - 20

1 children

Deletion Case 3

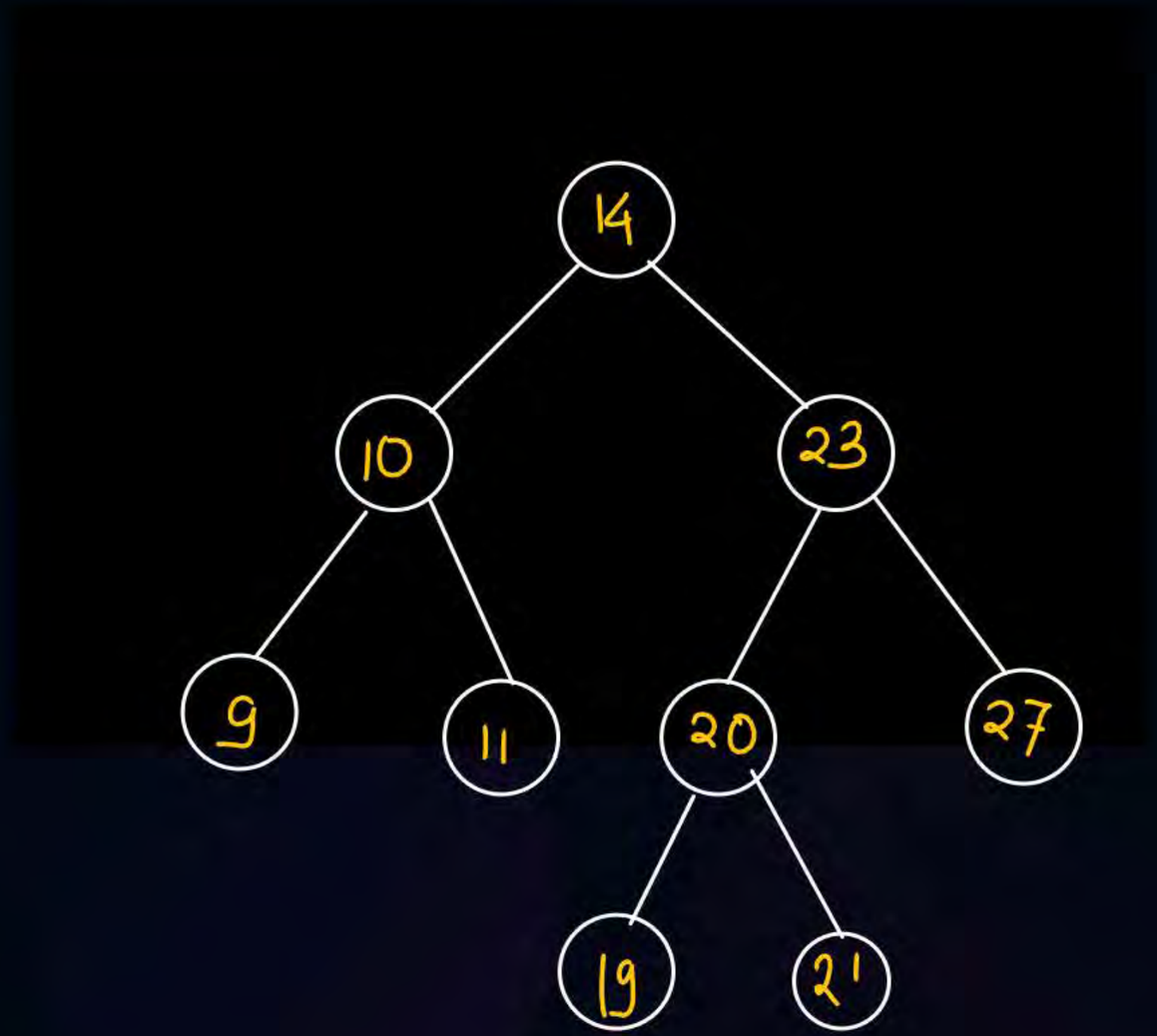
How many children a Successor/predecessor can have



14 predecessor - 11

14 Successor - 19

Deletion Case 3



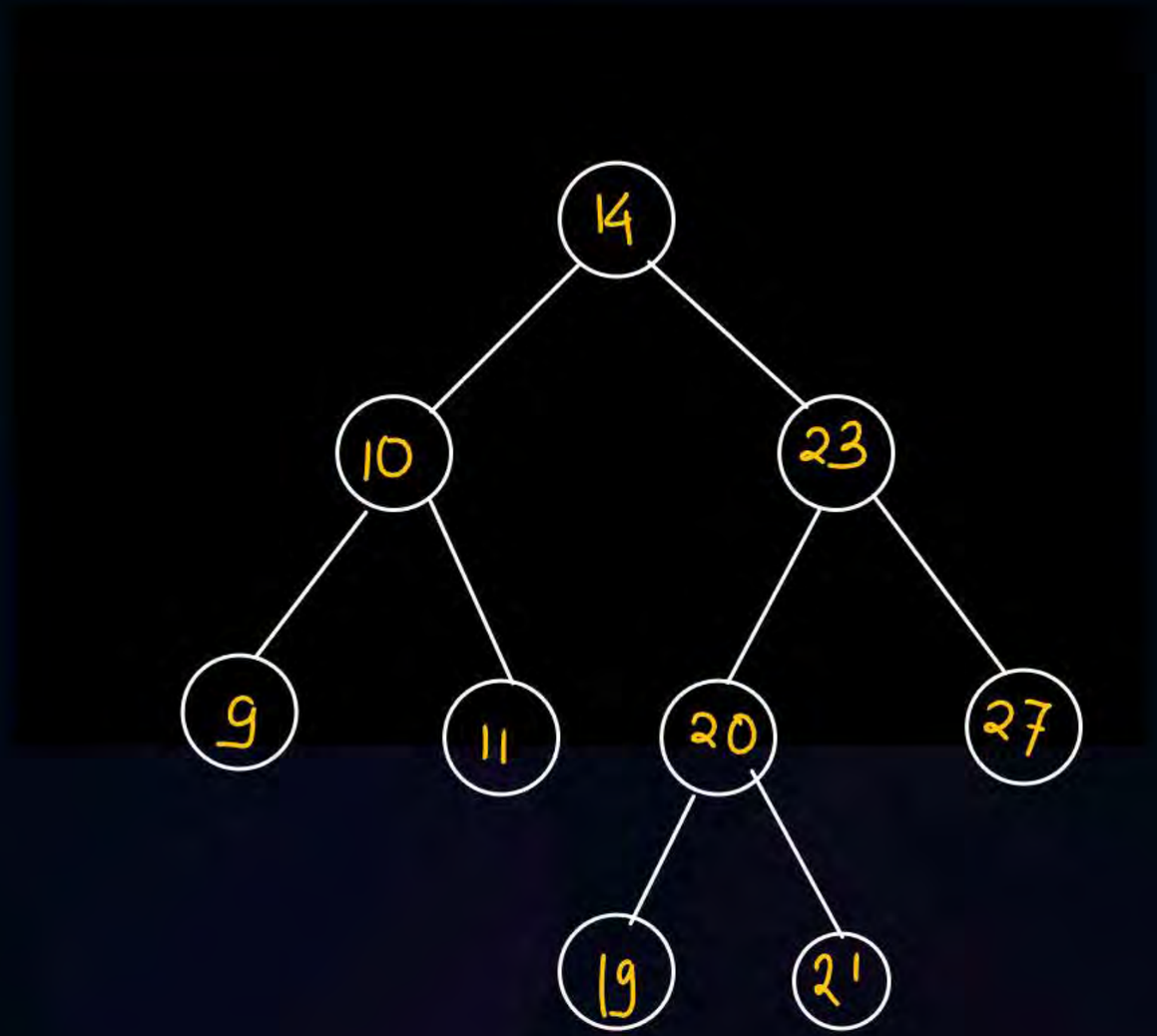
for a given node x

predecessor / Successor

can have 0 children

1 children (right child)

Deletion Case 3



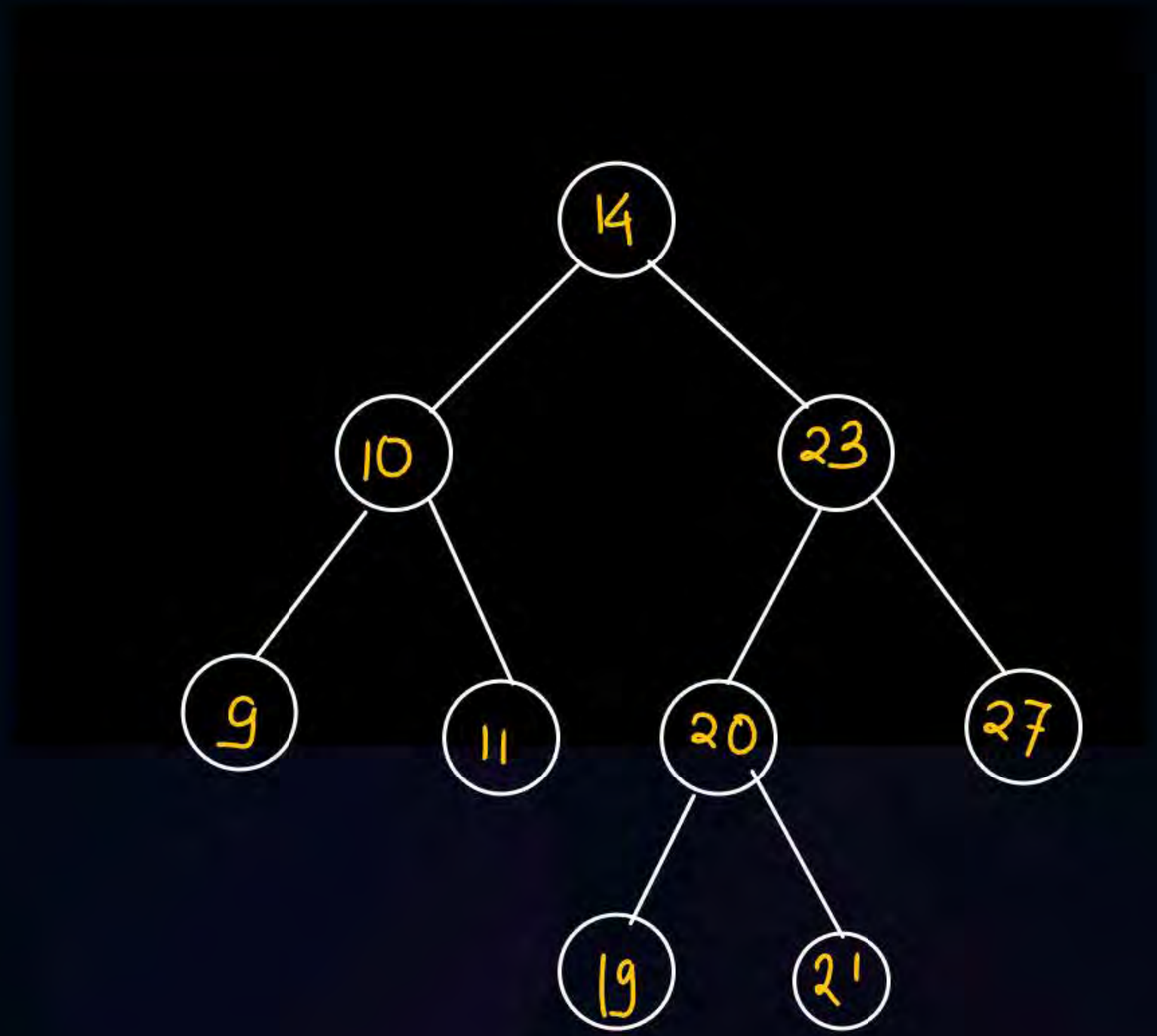
for a given node x

predecessor:

can have 0 children

1 children (Left child)

Deletion Case 3



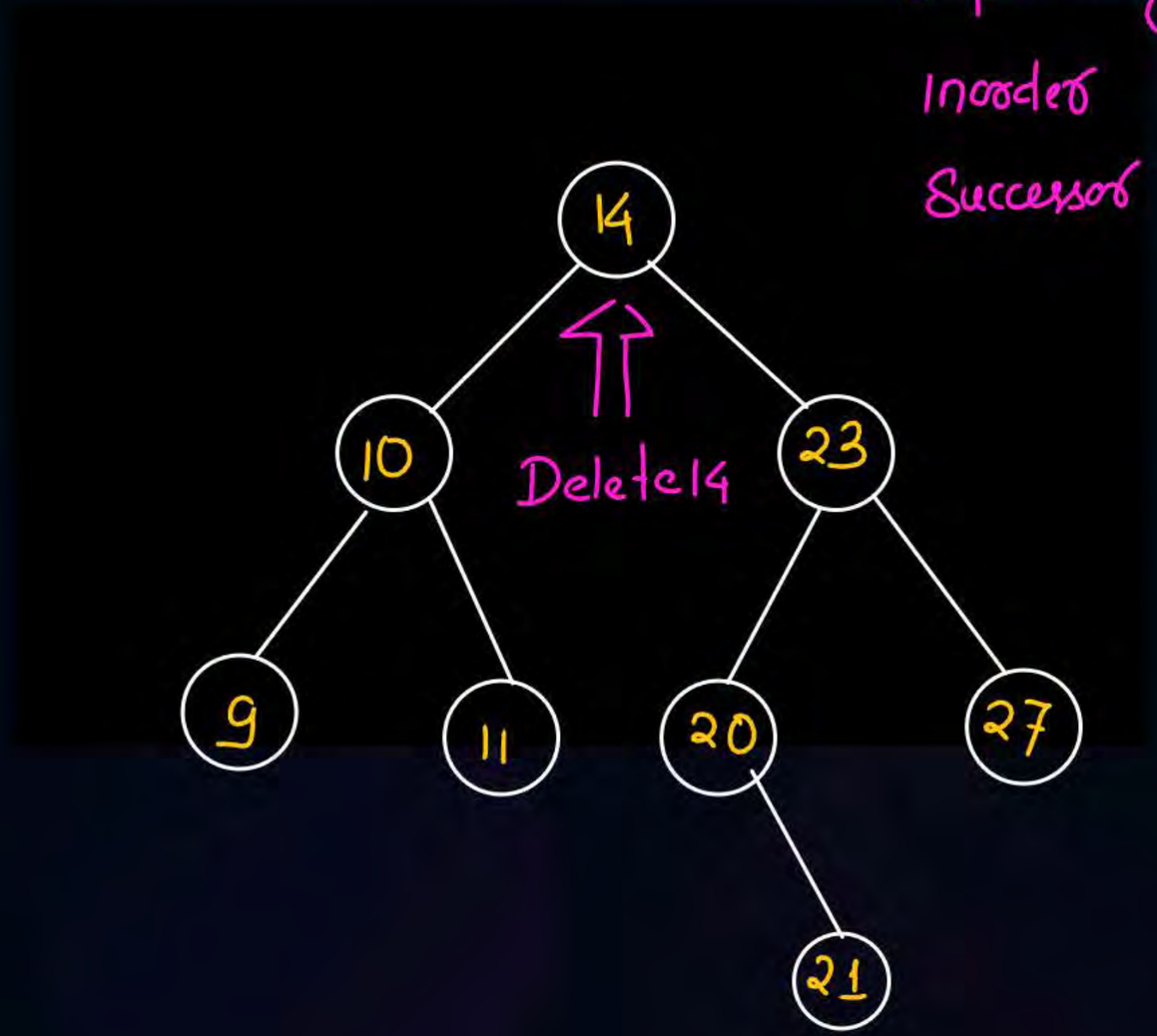
for a given node x

Successor

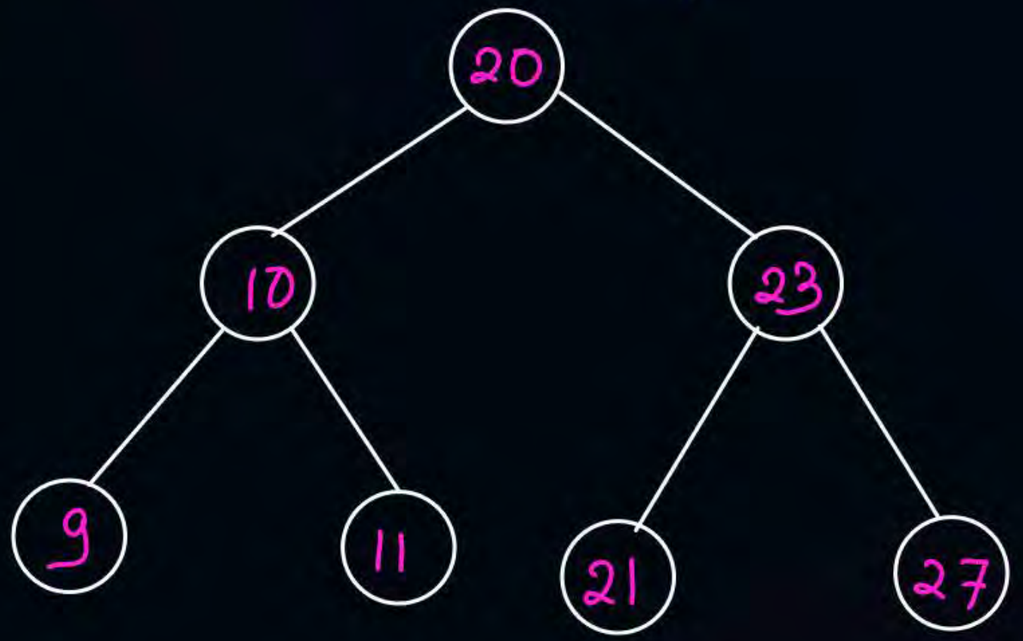
can have 0 children

1 children (right child)

Deletion Case 3

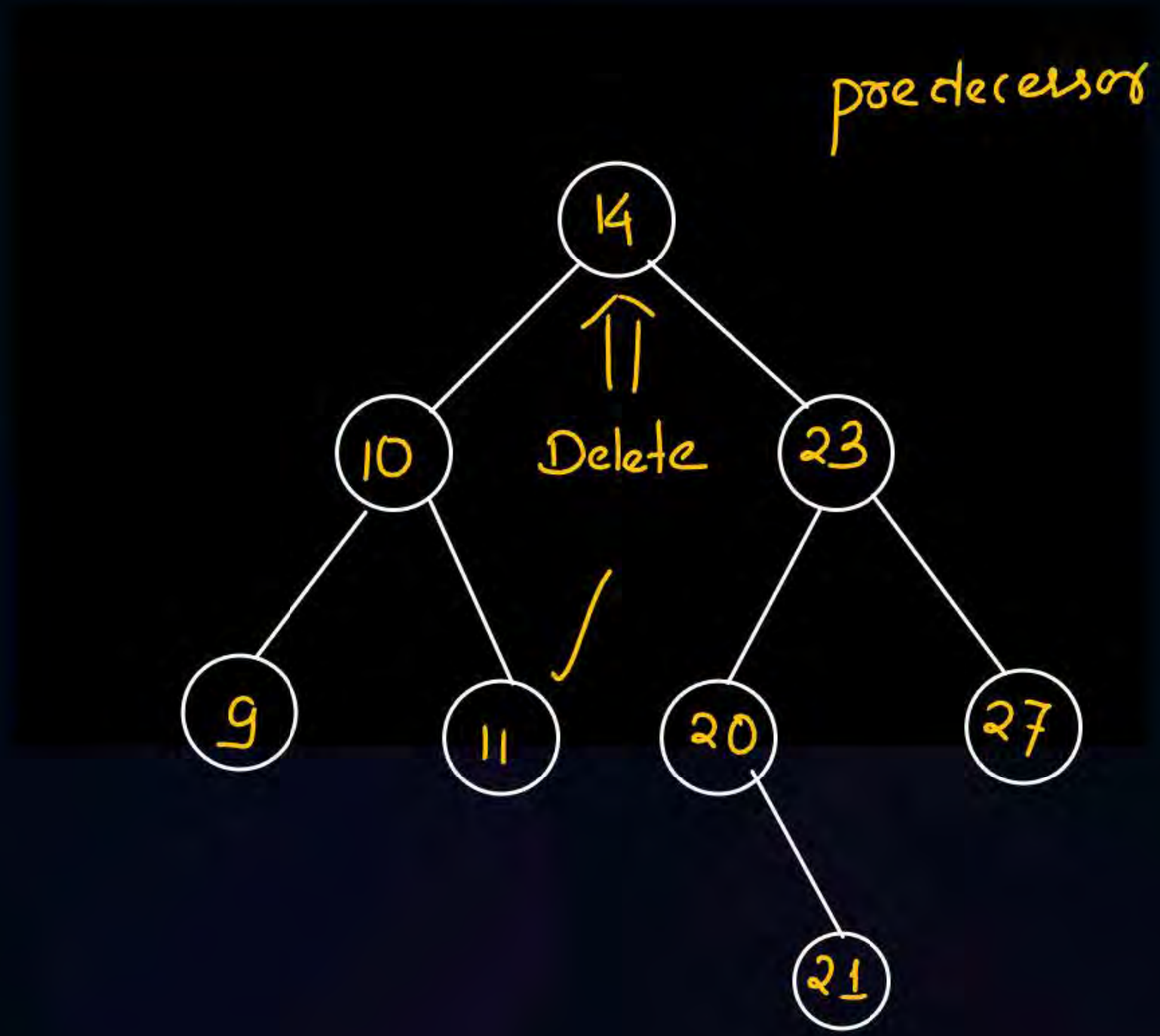


replaced by Inorder Successor



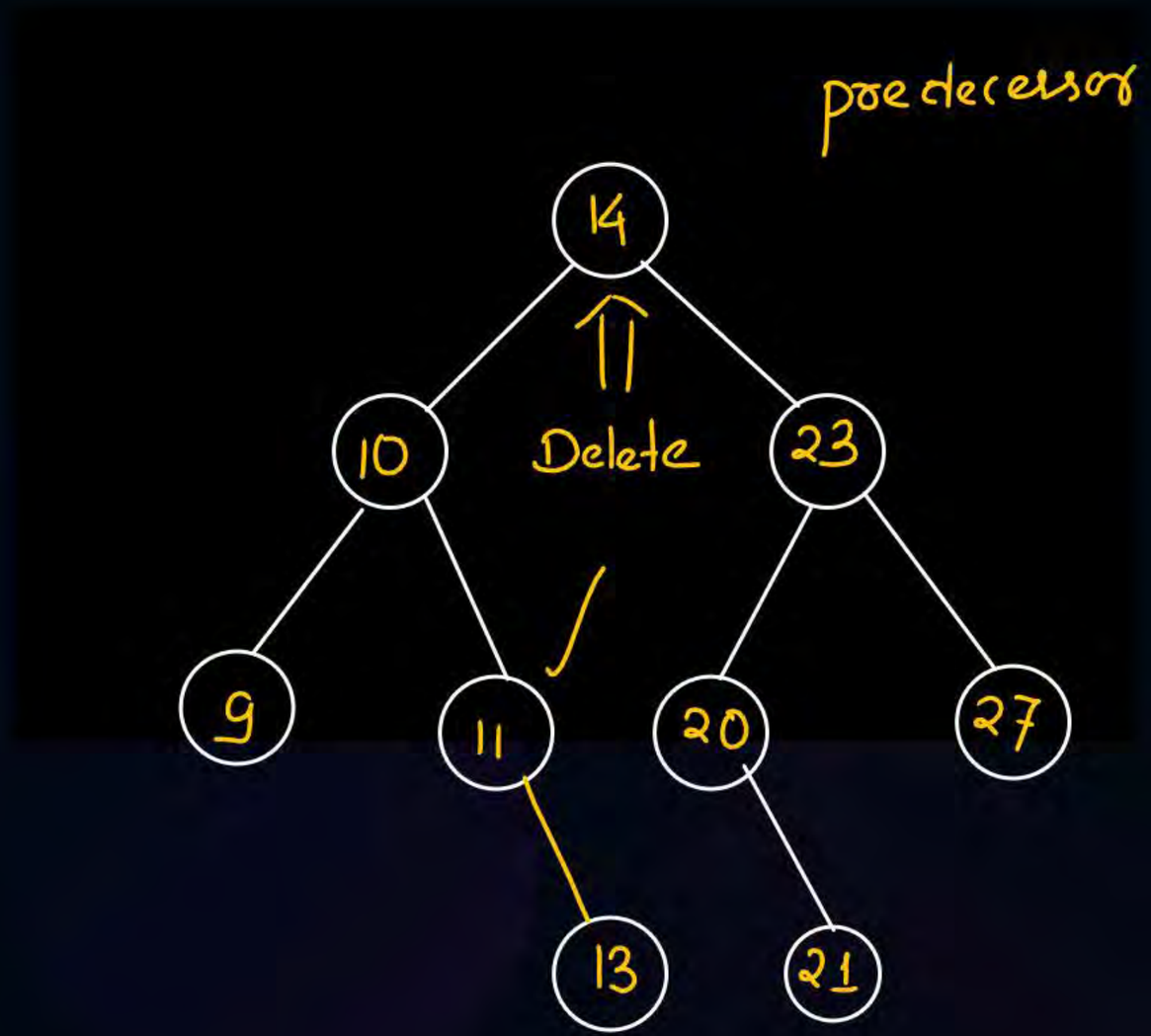
Children of 20 becomes Children of parent of 20 which is 23

Deletion Case 3



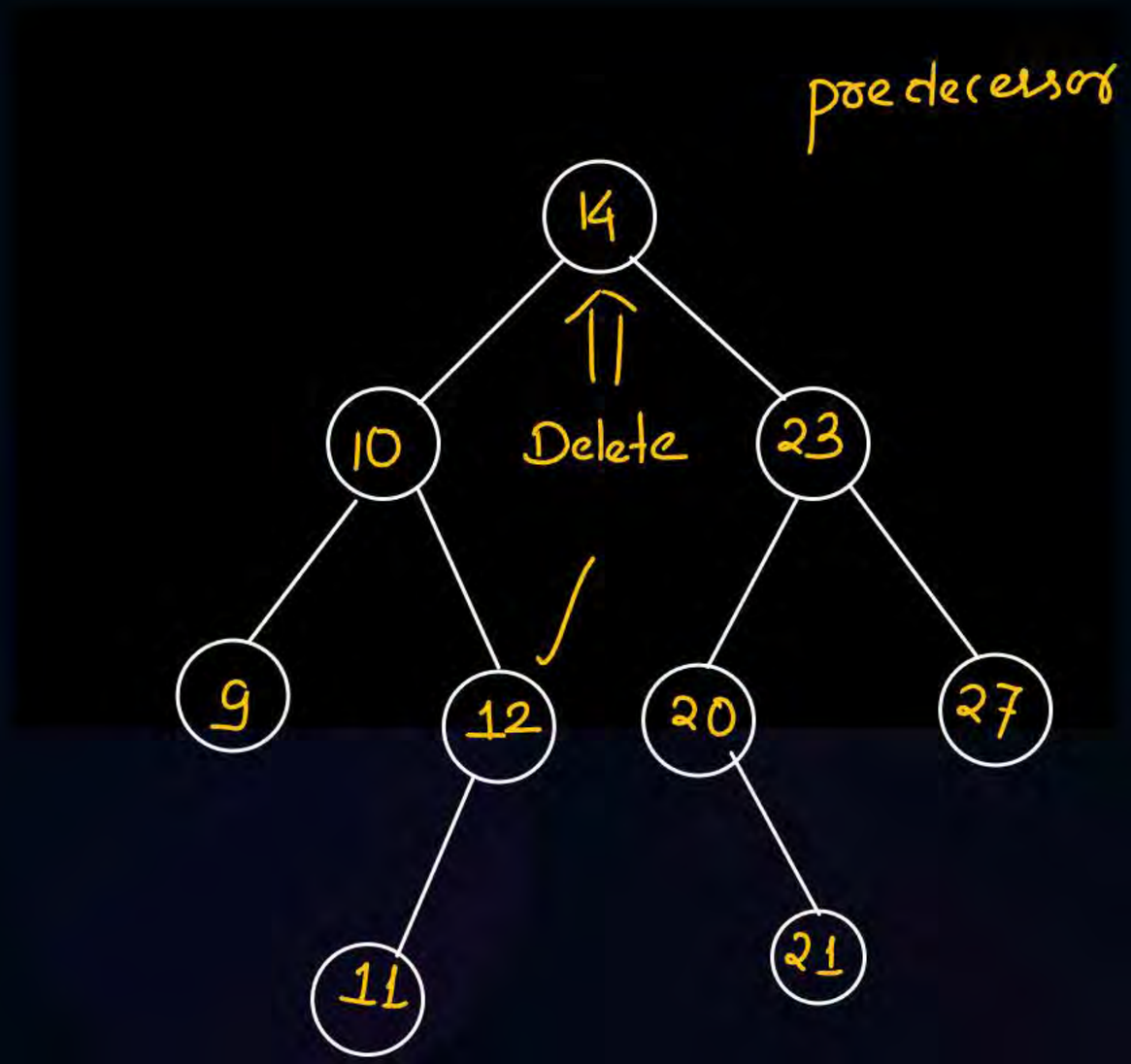


Deletion Case 3





Deletion Case 3





Topic : Binary Search Tree

Given an unlabeled tree with 3 nodes & 3 distinct keys. How many ways we can label it so that it becomes a BST ____

10, 11, 12



Slide

No. of BST possible with n distinct node is

Catalan No. $\frac{1}{n+1} {}^{2n}C_n$

HW. How many BST with 1,2,3,4 will have height : 3



Topic : AVL Tree

AVL Tree

AVL tree is

Height balanced tree

BST — Insert
Search
Min
Max
Delete

O(n)

Skewed tree can be formed.



Topic : AVL Tree

How Height balancing decided

The balancing factor of Node is calculated

for a given Node difference between Heigh in left

Subtree & right subtree

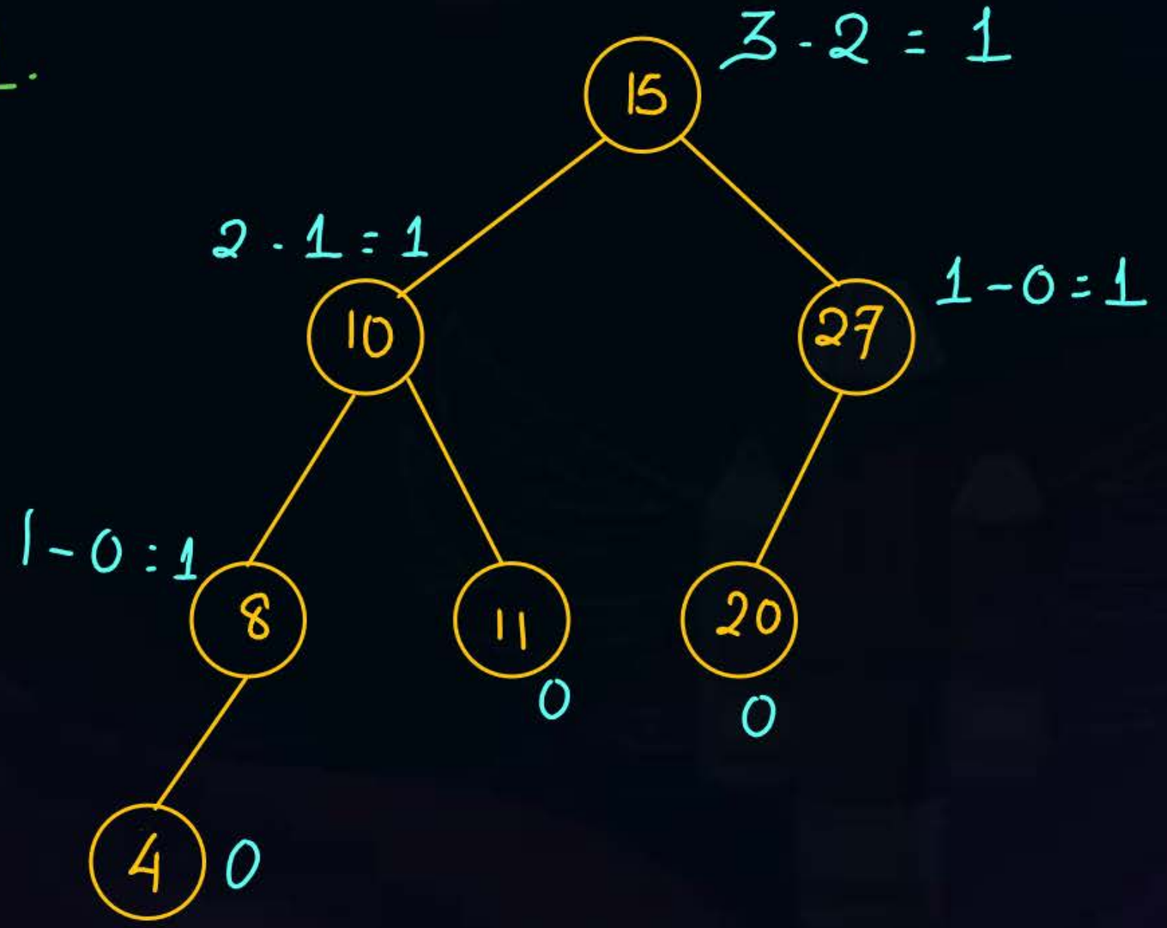
Height in Left subtree - Height in Right Subtree

Slide



Topic : AVL Tree

A given BST is an AVL tree if balancing factor of each Node is 0, 1 or -1.

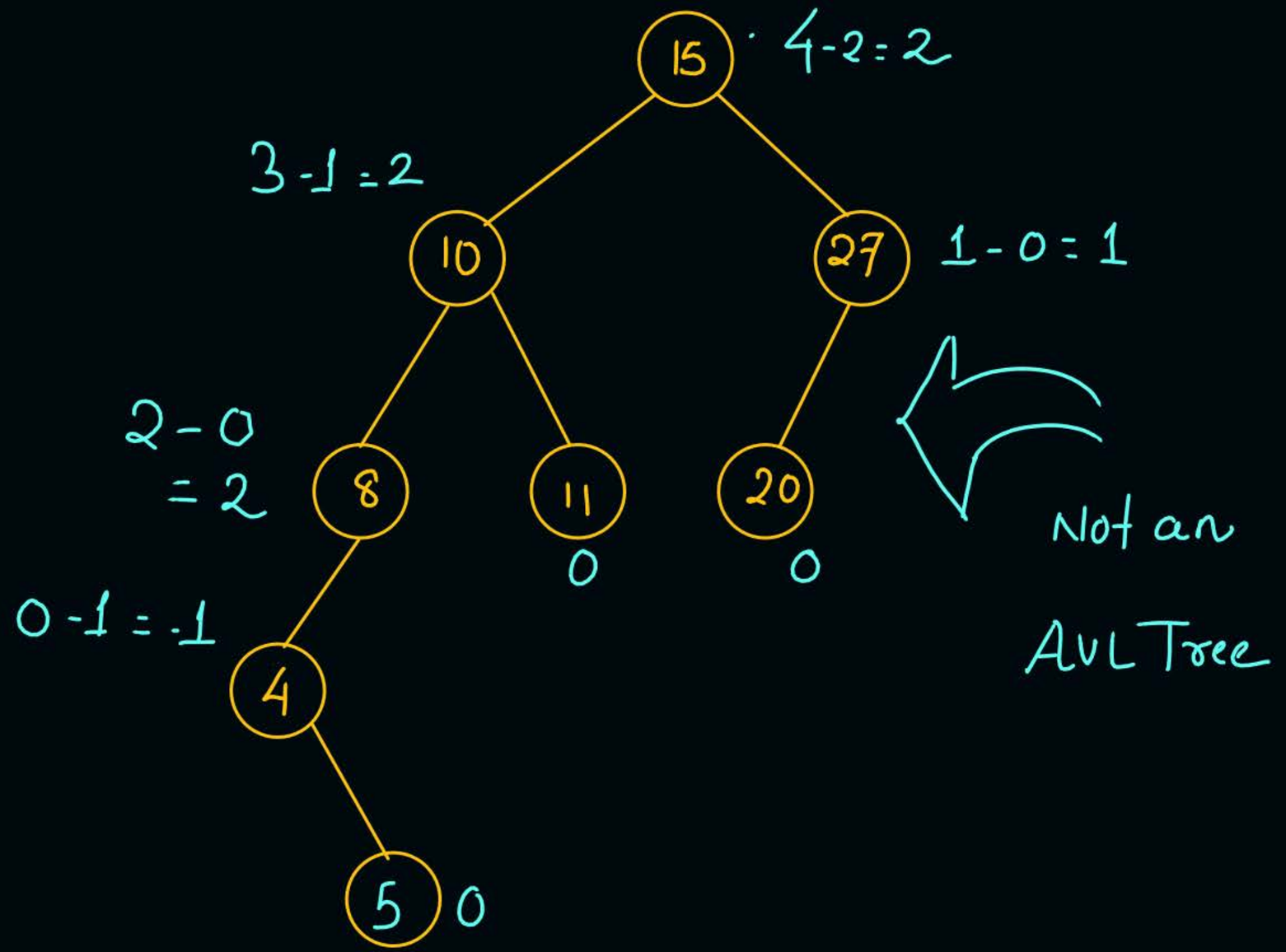


Slide

Insert 5

Critical Node

Identify

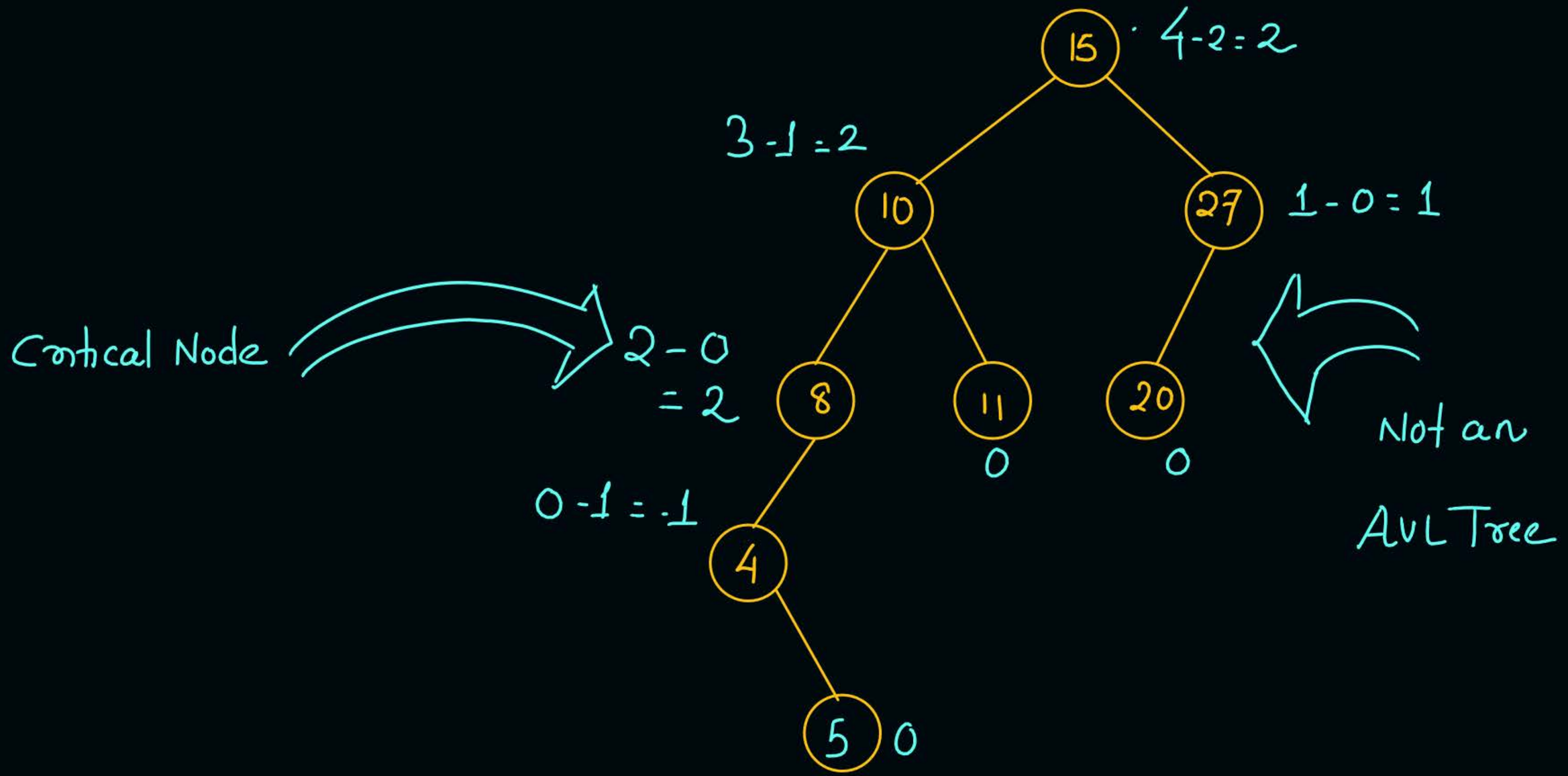




Topic : AVL Tree

from Newly inserted we move towards the root, and first Node that violates balancing property is critical Node.

Slide


15
4-2=2
3-1=2
10
27
1-0=1
Critical Node
2-0
=2
8
11
20
0
0
Not an
AVL Tree
0-1=-1
4
5
0



Topic : AVL Tree

for balancing we Need to perform Rotation

1 Single Rotation :-

- LL Rotation
- RR Rotation

2. Double Rotation :

- LR Rotation
- RL Rotation

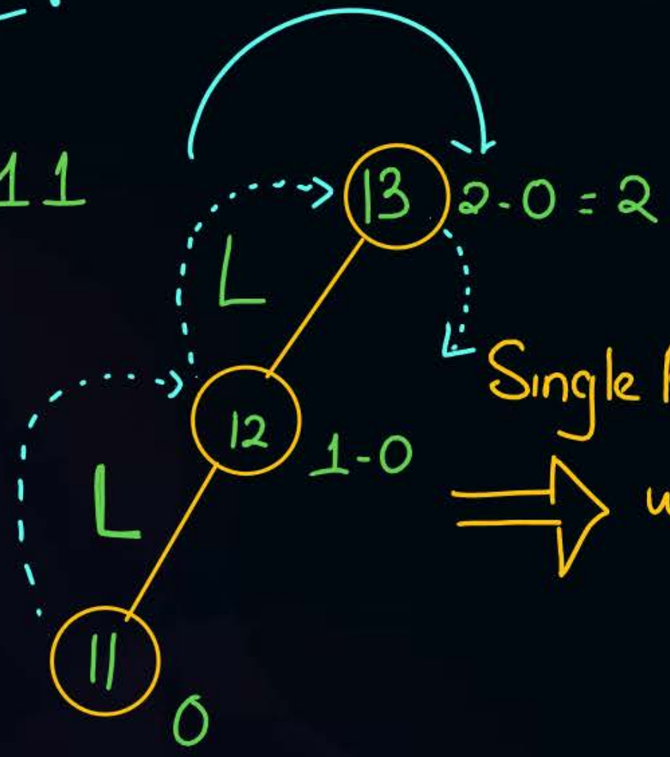
Slide



Topic : AVL Tree

LL Rotation :

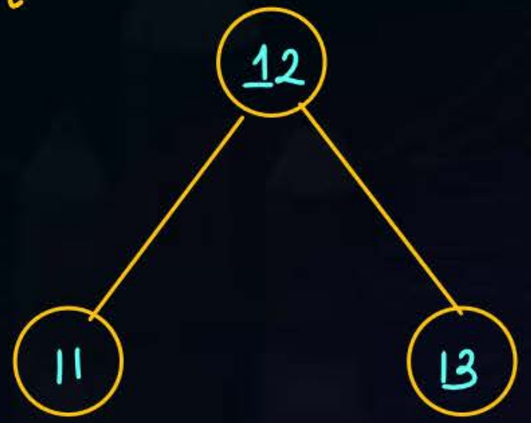
Insert 13, 12, 11



Critical Node 13

from critical Node to Newly inserted Node identify 2 edges

Single Right Rotation ⟹ will fix it



Slide



2 mins Summary

Topic — BST Deletion

Topic — AVL tree . LL Rotation

Topic — BST counting

Topic

Topic

Slide

THANK - YOU

Slide